UNIT 7

Forecasting hierarchical and grouped time series

By

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Forecasting hierarchical and grouped time series

- Forecasting hierarchical and grouped time series involves creating forecasts for data that is organized in a hierarchical or grouped structure. These types of data are common in industries like retail, finance, or any domain where data can be aggregated at different levels (e.g., sales data for products, stores, and regions).
- Hierarchical and grouped time series forecasting attempts to balance the need for accuracy at multiple levels of aggregation and disaggregation. Below, we break down the two types and discuss the key approaches.

Hierarchical time series

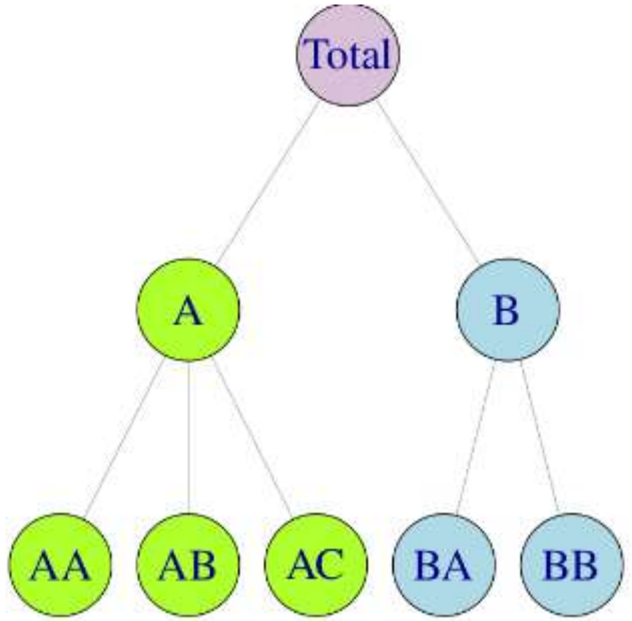
Figure shows a simple hierarchical structure.

At the top of the hierarchy is the “Total”, the most aggregate level of the data. The t_{th} observation of the Total series is denoted by y_t for $t=1,\ldots,T$.

The Total is disaggregated into two series, which in turn are divided into three and two series respectively at the bottom level of the hierarchy.

Below the top level, we use $y_{j,t}$ to denote the tth observation of the series corresponding to node j.

For example, $y_{A,t}$ denotes the tth observation of the series corresponding to node A, $y_{AB,t}$ denotes the tth observation of the series corresponding to node AB, and so on.



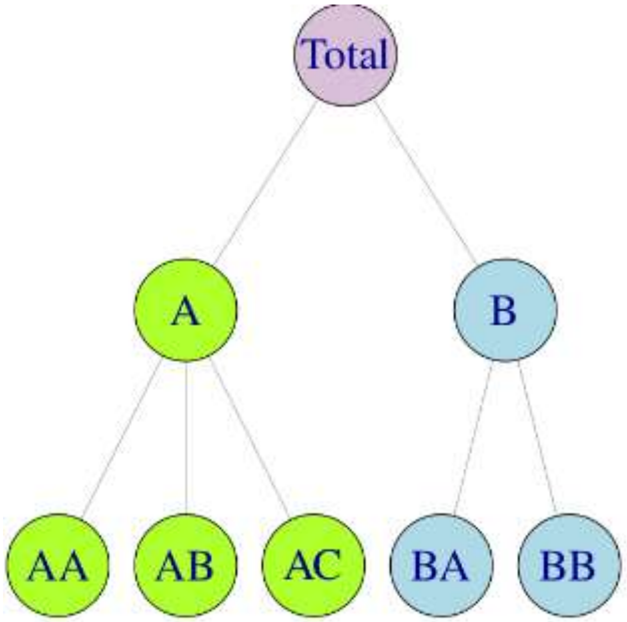
Hierarchical time series

- In this small example, the total number of series in the hierarchy is $n=1+2+5=8$, while the number of series at the bottom level is $m=5$. Note that $n>m$ in all hierarchies.
- For any time t, the observations at the bottom level of the hierarchy will sum to the observations of the series above. For example,

$$y_t = y_{\mathrm{AA},t} + y_{\mathrm{AB},t} + y_{\mathrm{AC},t} + y_{\mathrm{BA},t} + y_{\mathrm{BB},t},$$

$$y_{\mathrm{A},t} = y_{\mathrm{AA},t} + y_{\mathrm{AB},t} + y_{\mathrm{AC},t} \quad \text{and} \quad y_{\mathrm{B},t} = y_{\mathrm{BA},t} + y_{\mathrm{BB},t}.$$

$$y_t = y_{\mathrm{A},t} + y_{\mathrm{B},t}.$$



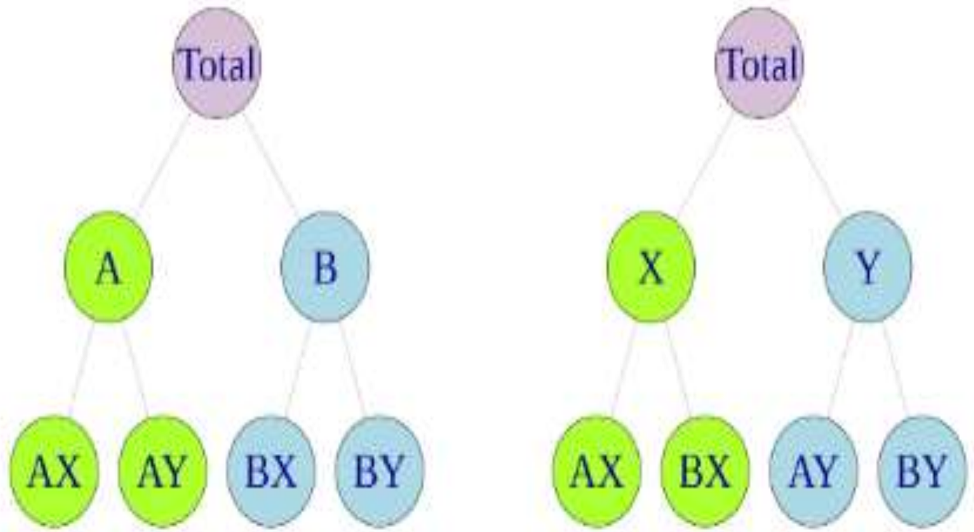
Grouped time series

- With grouped time series, the data structure does not naturally disaggregate in a unique hierarchical manner.
- Figure shows a simple grouped structure.

- At the top of the grouped structure is the Total, the most aggregate level of the data, again represented by y_t.

- The Total can be disaggregated by attributes (A, B) forming series $y_{A,t}$ and $y_{B,t}$, or by attributes (X, Y) forming series $y_{X,t}$ and $y_{Y,t}$.

- At the bottom level, the data are disaggregated by both attributes.



Grouped time series

- This example shows that there are alternative aggregation paths for grouped structures. For any time t, as with the hierarchical structure,

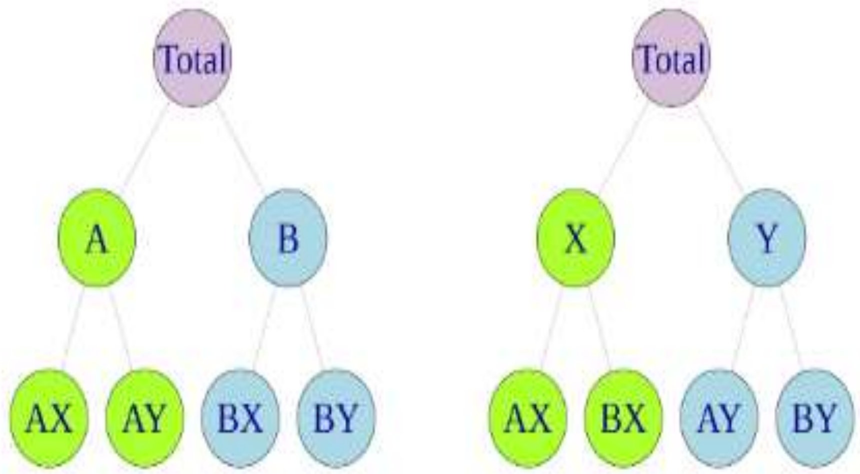
$$y_t = y_{AX,t} + y_{AY,t} + y_{BX,t} + y_{BY,t}.$$

However, for the first level of the grouped structure,

$$y_{A,t} = y_{AX,t} + y_{AY,t} \qquad y_{B,t} = y_{BX,t} + y_{BY,t}$$

but also

$$y_{X,t} = y_{AX,t} + y_{BX,t} \qquad y_{Y,t} = y_{AY,t} + y_{BY,t}.$$



Example

- **Example:** A company tracks its sales data. The hierarchical structure may look like:
- **Level 0:** Total sales
- **Level 1:** Sales by region (North, South, East, West)
- **Level 2:** Sales by store within each region
- **Level 3:** Sales by product within each store

Hierarchical Time Series

- 1. **Hierarchical Time Series**
- Hierarchical time series data is structured in a tree-like manner, where observations at one level can be aggregated to higher levels. The series is broken down into multiple levels of aggregation, such as:
- **Level 0 (Total)**: The most aggregated level (e.g., total sales for a company).
- **Level 1 (Groups)**: The series can be disaggregated into several groups (e.g., sales by region).
- **Level 2 (Subgroups)**: Each group can further be broken down into subgroups (e.g., sales by store).
- **Level 3 (Bottom level)**: The lowest level of the hierarchy (e.g., sales by product in each store).

Grouped Time Series

- **Grouped Time Series**
- Grouped time series refers to data that can be divided into different combinations based on cross-sections of factors (e.g., product categories, customer types) rather than a strict hierarchical tree structure.
- **Example:**
- **Level 0:** Total sales
- **Groupings:**
  - Sales by product category
  - Sales by customer type (retail/wholesale)
  - Sales by region (North/South)
  - Here, different combinations of the factors (product, customer type, region) form the "groups."

There are three primary approaches for achieving this:

1. Forecasting Hierarchical and Grouped Time Series
2. When dealing with these structures, the goal is to ensure that the forecasts at different levels are coherent. Coherence means that forecasts at the bottom level should sum up to match the forecasts at higher aggregation levels.

(A) Top-Down Approach

(B) Bottom-Up Approach

(C) Middle-Out Approach

Top-Down Approach

- This approach starts by forecasting the most aggregated level of the hierarchy and then disaggregating that forecast to lower levels using a set of proportions based on historical data or other rules.

Process:

- **Forecast the Total (Top-Level):** Forecast the most aggregated time series (e.g., total sales for all regions).
- **Allocate Downward:** Allocate the total forecast down to lower levels of the hierarchy (e.g., to regions, stores) based on proportions derived from historical data or other fixed rules.

Example:

- Let's say the company forecasts total sales for next month to be $100,000. Historical data shows that 40% of sales come from the North region, 30% from the South, 20% from the East, and 10% from the West. Using these proportions:
- Forecast for North = 40% of $100,000 = $40,000
- Forecast for South = 30% of $100,000 = $30,000
- Forecast for East = 20% of $100,000 = $20,000
- Forecast for West = 10% of $100,000 = $10,000

Advantages:

- Simple and easy to implement.
- Consistent at higher levels.

Disadvantages:

Bottom-Up Approach

- The bottom-up approach starts by forecasting the lowest level of the hierarchy and then aggregating these forecasts to higher levels.

Process:

- **Forecast Bottom Level**: First, generate forecasts for the most granular series (e.g., product sales in each store).
- **Aggregate Upward**: Aggregate the forecasts to form higher-level forecasts (e.g., region or total sales).
- **Example**:
- Assume we forecast product sales at each store:
- Store A: $5,000 (Product X), $3,000 (Product Y)
- Store B: $7,000 (Product X), $4,000 (Product Y)
- These can be summed up to get:
- Region 1 (Store A + Store B): $19,000

Total sales: $19,000

- **Advantages**:
- Bottom-up forecasting can capture nuances at the lower levels.
- High accuracy at the bottom levels.
- **Disadvantages**:
- Requires high-quality data at the most granular level. May lead to inconsistencies at higher levels if bottom-level forecasts are noisy.

Middle-Out Approach

- This is a hybrid method where you forecast at a middle level of the hierarchy and then either aggregate upwards or disaggregate downwards.

Process:

- **Choose a middle level** (e.g., region level).
- **Generate forecasts** at that level.
- **Aggregate upward** to the total and **disaggregate downward** to individual stores or products.

Example:

- Forecast sales at the region level and then:
- Aggregate up to get total sales.
- Disaggregate down to store or product level using proportions.

Advantages:

- Balances the simplicity of the top-down approach with the granularity of the bottom-up method.

Disadvantages:

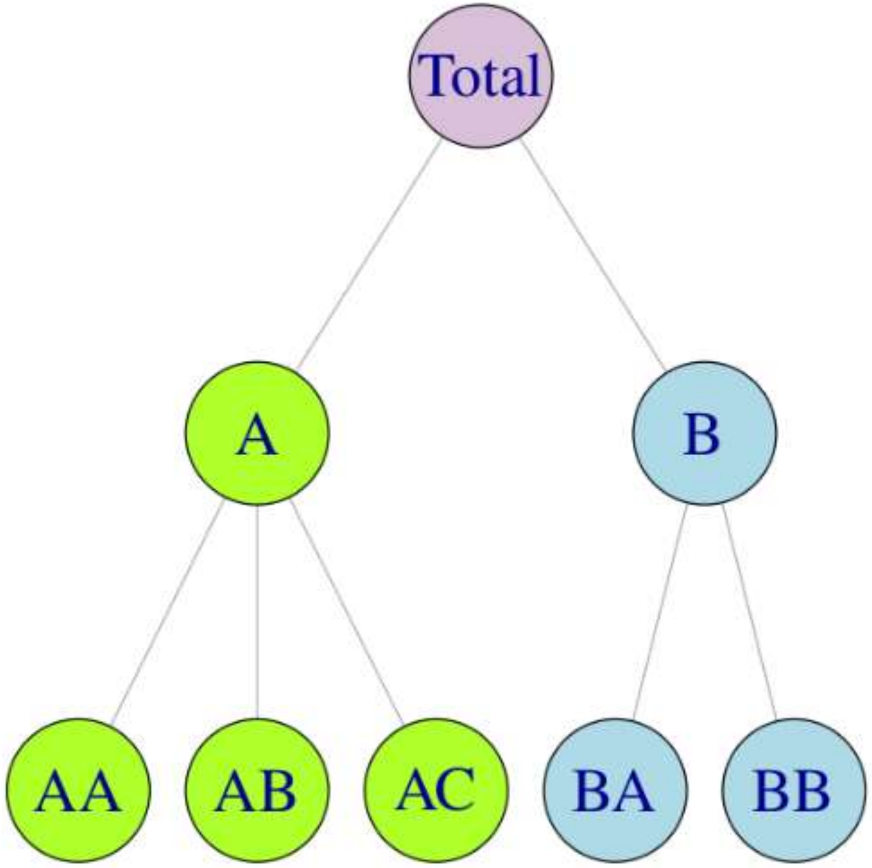
- May still introduce some inconsistencies depending on the middle level chosen.

- **Reconciliation Approaches**
- For both hierarchical and grouped time series, we often apply reconciliation techniques to ensure coherence across the levels. Reconciliation adjusts the forecasts so that the forecasts at different levels are consistent (i.e., lower-level forecasts sum to match higher-level forecasts). Some common reconciliation methods include:
- **OLS (Ordinary Least Squares) Reconciliation**: Minimizes forecast errors by adjusting forecasts at all levels using OLS.
- **MinT (Minimum Trace Reconciliation)**: Minimizes the total variance of the forecast errors across all levels, providing a better balance.

Example Application in Retail Sales Forecasting

- Consider a retail company that sells products across multiple regions, with stores located in each region. The company wants to forecast total sales, as well as sales by region, by store, and by product. Using hierarchical time series forecasting:
- **Top-Down Approach**: Forecast total sales and split them by region and store based on historical proportions.
- **Bottom-Up Approach**: Forecast sales at the store/product level and aggregate them to get regional and total forecasts.
- **Reconciliation**: Adjust the forecasts to ensure that the sum of store-level forecasts matches the regional and total sales forecasts.

EXAMPLE



- Let's break down this example in the image step by step:
- Structure Overview
- **Top-Level (Total):**
  - The top-most node is the **Total**, representing the aggregated forecast for the entire system (e.g., total sales, demand, or production for all entities).
- **Middle-Level (A and B):**
  - The next level consists of two main branches, **A** and **B**, representing two main categories or groups within the hierarchy.
  - For example, A and B could be sales in two different regions (Region A and Region B) or two departments of a business.
- **Bottom-Level (AA, AB, AC, BA, BB):**
  - At the lowest level, there are subgroups, such as **AA, AB, AC** under **A** and **BA, BB** under **B**. These represent the finest levels of disaggregation.
  - For example, **AA**, **AB**, and **AC** could be sales in different stores within Region A, while **BA** and **BB** could be sales in different

Top-Down Approach (From Total to the Bottom)

How It Works:

- In the **Top-Down** approach, you first generate a forecast for the top-level (Total) and then disaggregate it down to the lower levels (A, B, AA, AB, etc.) based on proportions derived from historical data.

Example:

- Suppose you forecast the **Total** sales to be $100,000 for the next period. You then use historical data to calculate what proportion of sales each level (A, B, AA, etc.) typically contributes.
  - Let's say historically, Region A (A) contributes 60% of the sales and Region B (B) contributes 40%.
  - So, forecast for A = 60% of $100,000 = $60,000, and for B = 40% of $100,000 = $40,000.
  - At the next level, you further break down the forecast for A:
  - If **AA** contributes 50% of Region A's sales, **AB** 30%, and **AC** 20%, then:
    - Forecast for AA = 50% of $60,000 = $30,000
    - Forecast for AB = 30% of $60,000 = $18,000
    - Forecast for AC = 20% of $60,000 = $12,000
    - Similarly, for Region B:
  - If **BA** contributes 70% of B's sales and **BB** 30%, then:
    - Forecast for BA = 70% of $40,000 = $28,000
    - Forecast for BB = 30% of $40,000 = $12,000

Advantages:

- Simple to implement.
- Ensures consistency at the top level.

Disadvantages:

Bottom-Up Approach (From Bottom to the Top)

How It Works:

- In the **Bottom-Up** approach, you start by forecasting at the lowest level of the hierarchy (AA, AB, AC, BA, BB) and then aggregate the forecasts upwards to generate forecasts for higher levels (A, B, and Total).

Example:

- Suppose you have accurate forecasts for:
  - **AA** = $30,000, **AB** = $18,000, **AC** = $12,000
  - **BA** = $28,000, **BB** = $12,000
  - You can then aggregate these bottom-level forecasts:
  - Forecast for A = AA + AB + AC = $30,000 + $18,000 + $12,000 = $60,000
  - Forecast for B = BA + BB = $28,000 + $12,000 = $40,000
  - Total Forecast = A + B = $60,000 + $40,000 = $100,000

Advantages:

- Ensures accuracy at the bottom level, which can be critical if you need detailed forecasts for individual stores or products.

Disadvantages:

- Can lead to inconsistencies at higher levels if bottom-level forecasts are noisy or inaccurate.

Middle-Out Approach (Combination of Both)

How It Works:

- The **Middle-Out** approach is a hybrid method where you forecast at a middle level (such as A and B) and then either aggregate upward or disaggregate downward.

Example:

- Suppose you forecast Region A and Region B:
  - Forecast for A = $60,000
  - Forecast for B = $40,000
  - You can aggregate these to get the **Total** forecast:
  - Total Forecast = A + B = $60,000 + $40,000 = $100,000
  - Then, you disaggregate the Region A forecast into AA, AB, and AC:
  - Forecast for AA = 50% of $60,000 = $30,000
  - Forecast for AB = 30% of $60,000 = $18,000
  - Forecast for AC = 20% of $60,000 = $12,000

  Advantages:

- Provides a balance between simplicity and accuracy.
- Allows for flexibility in focusing on important levels of the hierarchy.

Disadvantages:

- May still introduce some inconsistencies depending on which middle level is chosen.

Reconciliation Techniques

- Once the forecasts have been made using any of the above methods, reconciliation techniques are often applied to adjust forecasts across all levels of the hierarchy to ensure **coherence.** Coherence means that forecasts at lower levels should sum up correctly to match the forecasts at higher levels.
- Popular Reconciliation Techniques:
- **OLS (Ordinary Least Squares):** Minimizes forecast errors by adjusting forecasts at all levels using OLS regression.
- **MinT (Minimum Trace):** Minimizes the total variance of forecast errors across the hierarchy, providing a better-balanced reconciliation across levels.

Single level approaches

- Traditionally, forecasts of hierarchical or grouped time series involved selecting one level of aggregation and generating forecasts for that level. These are then either aggregated for higher levels, or disaggregated for lower levels, to obtain a set of coherent forecasts for the rest of the structure.

- The bottom-up approach

- A simple method for generating coherent forecasts is the “bottom-up” approach. This approach involves first generating forecasts for each series at the bottom level, and then summing these to produce forecasts for all the series in the structure.

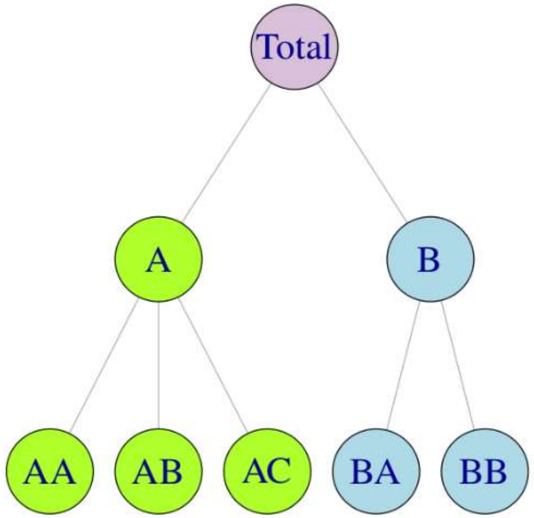
- For example, for the hierarchy of Figure , we first generate h-step-ahead forecasts for each of the bottom-level series:

$$\hat{y}_{\mathrm{AA},h},\ \hat{y}_{\mathrm{AB},h},\ \hat{y}_{\mathrm{AC},h},\ \hat{y}_{\mathrm{BA},h} \text{ and } \hat{y}_{\mathrm{BB},h}.$$

(We have simplified the previously used notation of $\hat{y}_{T+h|T}$ for brevity.)

Summing these, we get h-step-ahead coherent forecasts for the rest of the series:

$$\begin{aligned} \tilde{y}_h &= \hat{y}_{\mathrm{AA},h} + \hat{y}_{\mathrm{AB},h} + \hat{y}_{\mathrm{AC},h} + \hat{y}_{\mathrm{BA},h} + \hat{y}_{\mathrm{BB},h}, \\ \tilde{y}_{\mathrm{A},h} &= \hat{y}_{\mathrm{AA},h} + \hat{y}_{\mathrm{AB},h} + \hat{y}_{\mathrm{AC},h}, \\ \text{and} \quad \tilde{y}_{\mathrm{B},h} &= \hat{y}_{\mathrm{BA},h} + \hat{y}_{\mathrm{BB},h}. \end{aligned}$$



we will use the “tilde” notation to indicate coherent forecasts.

Top-down approaches

- Top-down approaches involve first generating forecasts for the Total series yt, and then disaggregating these down the hierarchy.
- Let p1,…,pm denote a set of disaggregation proportions which determine how the forecasts of the Total series are to be distributed to obtain forecasts for each series at the bottom level of the structure.
- For example, for the hierarchy of Figure, using proportions p1,…,p5 we get

$$\tilde{y}_{\mathrm{AA},t} = p_1\hat{y}_t, \quad \tilde{y}_{\mathrm{AB},t} = p_2\hat{y}_t, \quad \tilde{y}_{\mathrm{AC},t} = p_3\hat{y}_t, \quad \tilde{y}_{\mathrm{BA},t} = p_4\hat{y}_t \quad \text{and} \quad \tilde{y}_{\mathrm{BB},t} = p_5\hat{y}_t.$$

Once the bottom-level h-step-ahead forecasts have been generated, these are aggregated to generate coherent forecasts for the rest of the series.

There are several possible top-down methods that can be specified. The two most common top-down approaches specify disaggregation proportions based on the historical proportions of the data.

Average historical proportions

$$p_j = \frac{1}{T} \sum_{t=1}^{T} \frac{y_{j,t}}{y_t}$$

for $j = 1, \ldots, m$. Each proportion p_j reflects the average of the historical proportions of the bottom-level series $y_{j,t}$ over the period $t = 1, \ldots, T$ relative to the total aggregate y_t.

Proportions of the historical averages

$$p_j = \sum_{t=1}^{T} \frac{y_{j,t}}{T} \Big/ \sum_{t=1}^{T} \frac{y_t}{T}$$

for $j = 1, \ldots, m$. Each proportion p_j captures the average historical value of the bottom-level series $y_{j,t}$ relative to the average value of the total aggregate y_t.

- A convenient attribute of such top-down approaches is their simplicity. One only needs to model and generate forecasts for the most aggregated top-level series.
- In general, these approaches seem to produce quite reliable forecasts for the aggregate levels and they are useful with low count data.
- On the other hand, one disadvantage is the loss of information due to aggregation.
- Using such top-down approaches, we are unable to capture and take advantage of individual series characteristics such as time dynamics, special events, different seasonal patterns, etc.

Forecast proportions

- Consider a one level hierarchy.
- We first generate h-step-ahead forecasts for all of the series.
- We don’t use these forecasts directly, and they are not coherent (they don’t add up correctly).
- Let’s call these “initial” forecasts.
- We calculate the proportion of each h-step-ahead initial forecast at the bottom level, to the aggregate of all the h-step-ahead initial forecasts at this level.
- We refer to these as the forecast proportions, and we use them to disaggregate the top-level h-step-ahead initial forecast in order to generate coherent forecasts for the whole of the hierarchy.

Hierarchical Example:

- Let's say we have a company that sells products in two regions: **North** and **South**. The hierarchy lo

Step 1: **Generate Initial Forecasts (Non-coherent)**

Suppose we generate **initial h-step-ahead forecasts** for each region, but these forecasts are **not coherent** — meaning they don't add up to the total sales forecast.

Let's say the initial forecasts for each region are:

- Initial Forecast for **North** = 500 units
- Initial Forecast for **South** = 400 units
- The **Total Sales** forecast (sum of North and South) = 500 + 400 = 900 units

Now, let's say the **Top-level Total Sales Forecast** (company-wide forecast) is **1000 units** instead of 900 (which means the initial regional forecasts do not sum to this total and are not coherent).

Step 2: **Calculate Forecast Proportions**

Next, we calculate the **forecast proportions** for each region, based on the **initial forecasts**.

$$\text{Forecast Proportion for North} = \frac{500}{500+400} = \frac{500}{900} = 0.5556$$

$$\text{Forecast Proportion for South} = \frac{400}{500+400} = \frac{400}{900} = 0.4444$$

Hierarchical Example:

Step 3: **Disaggregate the Top-Level Forecast**

Now, we distribute the **Top-level Total Sales Forecast** (1000 units) proportionally to each region, using the forecast proportions calculated above.

- For **North**, the coherent forecast will be:

$$\text{Coherent Forecast for North} = 1000 \times 0.5556 = 555.6 \text{ units}$$

- For **South**, the coherent forecast will be:

$$\text{Coherent Forecast for South} = 1000 \times 0.4444 = 444.4 \text{ units}$$

Result: **Coherent Forecasts**

Now the forecasts for the regions **add up correctly** to the top-level forecast:

- **North**: 555.6 units
- **South**: 444.4 units

The **Total** (North + South) = 555.6 + 444.4 = **1000 units**, which matches the top-level total sales forecast.

Forecast proportions

For a K-level hierarchy, this process is repeated for each node, going from the top to the bottom level. Applying this process leads to the following general rule for obtaining the forecast proportions:

$$p_j = \prod_{\ell=0}^{K-1} \frac{\hat{y}_{j,h}^{(\ell)}}{\hat{S}_{j,h}^{(\ell+1)}}$$

where $j = 1, 2, \dots, m$, $\hat{y}_{j,h}^{(\ell)}$ is the h-step-ahead initial forecast of the series that corresponds to the node which is ℓ levels above j, and $\hat{S}_{j,h}^{(\ell)}$ is the sum of the h-step-ahead initial forecasts below the node that is ℓ levels above node j and are directly connected to that node. These forecast proportions disaggregate the h-step-ahead initial forecast of the Total series to get h-step-ahead coherent forecasts of the bottom-level series.

Hierarchical Structure:

- **Level 0**: Total
- **Level 1**: Group A and Group B
- **Level 2**: Subgroups under A (AA, AB, AC) and under B (BA, BB)

Step 1: Non-Coherent Initial Forecasts

Let's assume we have the following non-coherent forecasts:

- **Total Forecast**:
  $\hat{Y}_{\text{Total},t} = 1000$
- **Level 1 (Groups A and B)**:
  $\hat{Y}_{A,t} = 650$
  $\hat{Y}_{B,t} = 400$

  (Note: The sum of these forecasts is $650 + 400 = 1050$, which exceeds the Total forecast of 1000. Hence, they are non-coherent.)

- Level 2 (Subgroups under A and B):
  - Group A:
    $\hat{Y}_{AA,t} = 300$
    $\hat{Y}_{AB,t} = 250$
    $\hat{Y}_{AC,t} = 150$

    (Sum = $300 + 250 + 150 = 700$, which does not match Group A's forecast of 650.)
  - Group B:
    $\hat{Y}_{BA,t} = 200$
    $\hat{Y}_{BB,t} = 250$

    (Sum = $200 + 250 = 450$, which does not match Group B's forecast of 400.)

Step 2: Proportion Calculation to Achieve Coherence

1. Adjust Level 1 Forecasts (Group A and B)

To make the forecasts coherent, we need to redistribute the forecasts for Groups A and B based on their initial proportions.

Calculate Proportions for Level 1:

- The total initial forecast for Level 1 is $650 + 400 = 1050$.

Proportions:

$$P_A = \frac{\hat{Y}_{A,t}}{1050} = \frac{650}{1050} \approx 0.619$$

$$P_B = \frac{\hat{Y}_{B,t}}{1050} = \frac{400}{1050} \approx 0.381$$

Adjust Forecasts: Now, we adjust the forecasts for Group A and Group B to match the Total forecast of 1000:

$$\hat{Y}_{A,t} = P_A \times \hat{Y}_{\text{Total},t} = 0.619 \times 1000 \approx 619$$

$$\hat{Y}_{B,t} = P_B \times \hat{Y}_{\text{Total},t} = 0.381 \times 1000 \approx 381$$

2. **Adjust Level 2 Forecasts (Subgroups under A and B)**

For Group A (AA, AB, AC):

Initial Sum for Group A: 700

Calculate Proportions:

$$P_{AA} = \frac{\hat{Y}_{AA,t}}{700} = \frac{300}{700} \approx 0.429$$

$$P_{AB} = \frac{\hat{Y}_{AB,t}}{700} = \frac{250}{700} \approx 0.357$$

$$P_{AC} = \frac{\hat{Y}_{AC,t}}{700} = \frac{150}{700} \approx 0.214$$

Adjust Forecasts for Subgroups: Now, we adjust the forecasts based on the new forecast for Group A (619):

$$\hat{Y}_{AA,t} = P_{AA} \times \hat{Y}_{A,t} = 0.429 \times 619 \approx 265.57 \approx 266$$

$$\hat{Y}_{AB,t} = P_{AB} \times \hat{Y}_{A,t} = 0.357 \times 619 \approx 220.57 \approx 221$$

$$\hat{Y}_{AC,t} = P_{AC} \times \hat{Y}_{A,t} = 0.214 \times 619 \approx 132.86 \approx 133$$

Check Sum for Group A:

$$\hat{Y}_{AA,t} + \hat{Y}_{AB,t} + \hat{Y}_{AC,t} = 266 + 221 + 133 = 620$$

For Group B (BA, BB):

Initial Sum for Group B: 450

Calculate Proportions:

$$P_{BA} = \frac{\hat{Y}_{BA,t}}{450} = \frac{200}{450} \approx 0.444$$

$$P_{BB} = \frac{\hat{Y}_{BB,t}}{450} = \frac{250}{450} \approx 0.556$$

Adjust Forecasts for Subgroups: Now, we adjust the forecasts based on the new forecast for Group B (381):

$$\hat{Y}_{BA,t} = P_{BA} \times \hat{Y}_{B,t} = 0.444 \times 381 \approx 169.524 \approx 170$$

$$\hat{Y}_{BB,t} = P_{BB} \times \hat{Y}_{B,t} = 0.556 \times 381 \approx 211.476 \approx 211$$

Check Sum for Group B:

$$\hat{Y}_{BA,t} + \hat{Y}_{BB,t} = 170 + 211 = 381$$

Final Coherent Forecasts

- Total Forecast: 1000
- Group A Forecast: 619
  - Subgroups: $\hat{Y}_{AA,t} = 266, \hat{Y}_{AB,t} = 221, \hat{Y}_{AC,t} = 133$
- Group B Forecast: 381
  - Subgroups: $\hat{Y}_{BA,t} = 170, \hat{Y}_{BB,t} = 211$

Forecast reconciliation
Matrix notation

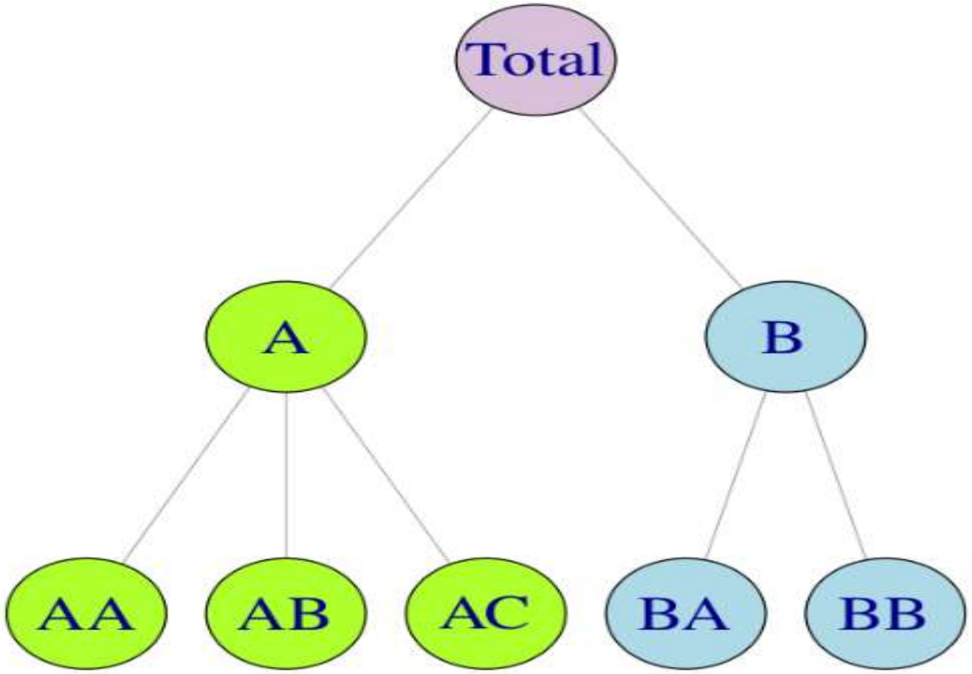
$$y_t = y_{AA,t} + y_{AB,t} + y_{AC,t} + y_{BA,t} + y_{BB,t},$$

Hierarchical structure

$$y_{A,t} = y_{AA,t} + y_{AB,t} + y_{AC,t} \quad \text{and} \quad y_{B,t} = y_{BA,t} + y_{BB,t}.$$

$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{AA,t} \\ y_{AB,t} \\ y_{AC,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AA,t} \\ y_{AB,t} \\ y_{AC,t} \\ y_{BA,t} \\ y_{BB,t} \end{bmatrix}$$

$$y_t = y_{A,t} + y_{B,t}$$



or in more compact notation

yt=Sbt,

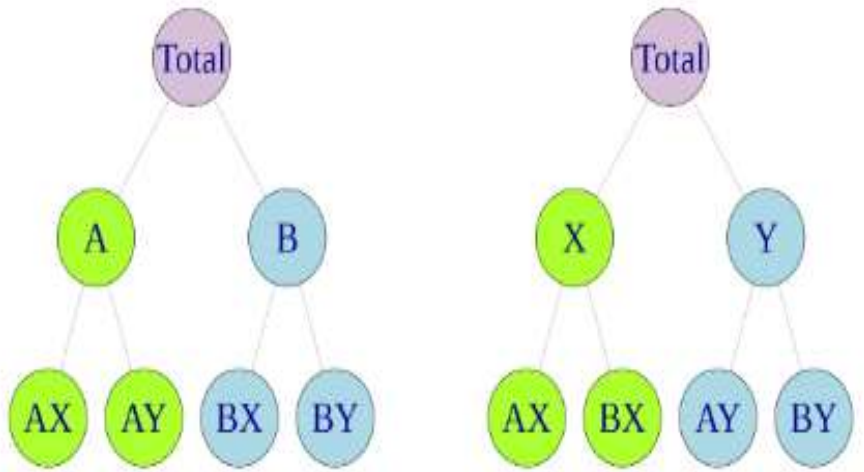
Forecast reconciliation Matrix notation

(Grouped structure)

- Similarly for the grouped structure of Figure we write

$$\begin{bmatrix} y_t \\ y_{A,t} \\ y_{B,t} \\ y_{X,t} \\ y_{Y,t} \\ y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_{AX,t} \\ y_{AY,t} \\ y_{BX,t} \\ y_{BY,t} \end{bmatrix},$$

$$\boldsymbol{y}_t = \boldsymbol{S}\boldsymbol{b}_t,$$



$$y_t = y_{AX,t} + y_{AY,t} + y_{BX,t} + y_{BY,t}.$$

However, for the first level of the grouped structure,

$$y_{A,t} = y_{AX,t} + y_{AY,t} \qquad y_{B,t} = y_{BX,t} + y_{BY,t}$$

but also

$$y_{X,t} = y_{AX,t} + y_{BX,t} \qquad y_{Y,t} = y_{AY,t} + y_{BY,t}.$$

Forecast reconciliation

- Forecast reconciliation in time series refers to the process of ensuring that forecasts made at different levels of a hierarchical or grouped structure are consistent with one another. In practical terms, it means adjusting forecasts for various subcomponents (such as products, regions, or time horizons) so that they "add up" to match the forecasts for higher levels of the hierarchy.
- Why Forecast Reconciliation is Needed:
- **Improved Accuracy:** Reconciliation ensures that the forecasts for various components are consistent and improves the overall forecast accuracy by integrating information across the hierarchy.
- **Business Consistency:** In many cases, organizations require consistency across departments or product lines, where disaggregated forecasts must roll up to the overall forecast.

Mapping matrices

All coherent forecasting approaches for either hierarchical or grouped structures can be represented as

$$\tilde{\boldsymbol{y}}_h = \boldsymbol{S}\boldsymbol{G}\hat{\boldsymbol{y}}_h,$$

where $\boldsymbol{G}$ is a matrix that maps the base forecasts into the bottom level, and the summing matrix $\boldsymbol{S}$ sums these up using the aggregation structure to produce a set of **coherent forecasts** $\tilde{\boldsymbol{y}}_h$.

The G matrix is defined according to the approach implemented. For example if the bottom-up approach is used to forecast the hierarchy

$$\boldsymbol{G} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

If any of the top-down approaches were used then

$$\boldsymbol{G} = \begin{bmatrix} p_1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ p_5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Forecast reconciliation

- The traditional methods considered so far are limited in that they only use base forecasts from a single level of aggregation which have either been aggregated or disaggregated to obtain forecasts at all other levels.
- Hence, they use limited information. However, in general, we could use other G matrices, and then SG combines and reconciles all the base forecasts in order to produce coherent forecasts.

The MinT optimal reconciliation approach

- Wickramasuriya et al. (2019) found a G matrix that minimises the total forecast variance of the set of coherent forecasts, leading to the MinT (Minimum Trace) optimal reconciliation approach.
- Suppose we generate coherent forecasts using Equation. First we want to make sure we have unbiased forecasts. If the base forecasts yh are unbiased, then the coherent forecasts yh will be unbiased provided

SGS=S

- This provides a constraint on the matrix G.

The MinT optimal reconciliation approach

Next we need to find the errors in our forecasts. Wickramasuriya et al. (2019) show that the variance-covariance matrix of the h-step-ahead coherent forecast errors is given by

$$\boldsymbol{V}_h = \mathrm{Var}[\boldsymbol{y}_{T+h} - \tilde{\boldsymbol{y}}_h] = \boldsymbol{S}\boldsymbol{G}\boldsymbol{W}_h\boldsymbol{G}'\boldsymbol{S}'$$

where $\boldsymbol{W}_h = \mathrm{Var}[(\boldsymbol{y}_{T+h} - \hat{\boldsymbol{y}}_h)]$ is the variance-covariance matrix of the corresponding base forecast errors.

The objective is to find a matrix $\boldsymbol{G}$ that minimises the error variances of the coherent forecasts. These error variances are on the diagonal of the matrix $\boldsymbol{V}_h$, and so the sum of all the error variances is given by the trace of the matrix $\boldsymbol{V}_h$. Wickramasuriya et al. (2019) show that the matrix $\boldsymbol{G}$ which minimises the trace of $\boldsymbol{V}_h$ such that $\boldsymbol{S}\boldsymbol{G}\boldsymbol{S} = \boldsymbol{S}$, is given by

$$\boldsymbol{G} = (\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}.$$

The MinT optimal reconciliation approach

Therefore, the optimally reconciled forecasts are given by

$$\tilde{\boldsymbol{y}}_h = \boldsymbol{S}(\boldsymbol{S}'\boldsymbol{W}_h^{-1}\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{W}_h^{-1}\hat{\boldsymbol{y}}_h.$$